Comparison Between Methods for Evaluating DIR Quality in Lungs

Ryan Andosca, MS[1], Peter Boyle, MS[1], Michael Lauria, PhD[1], Claudia R. Miller, BS[1], Louise Naumann[1], Minji Kim[1], Dylan P. O'Connell[1], Ricky Savjani, MD, PhD[2], Drew Moghanaki, MD, MPH[2] and Daniel A. Low, PhD[1]

(1)Department of Radiation Oncology, University of California, Los Angeles, Los Angeles, CA

(2)University of California, Los Angeles, Los Angeles

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INTRODUCTION

Deformable image registration is a key component of many free-breathing lung function studies, including breathing motion modeling for radiation therapy. DIR often fails locally while providing an overall good registration and quantifying and localizing those failure regions is time consuming and qualitative. Our proposed methods may provide for efficient detection of poor overall registration quality.

AIM

Image registration poses a difficult problem for applications that use multi-scan lung imaging modalities such as 5DCT. Image registration quality drives the application's accuracy, so measuring registration quality is important, though often time-consuming. We compare several metrics for evaluating overall image registration quality: average absolute HU-difference, average absolute DVF curl, percent lung volume above a curl threshold, percent lung volume within an HU-difference range, and a manual evaluation of registration quality. We then use the deeds registration algorithm and a deeds with follow-up pTV registration on a large group of patients and observe the absolute HU difference for each method.

METHOD

Initial Analysis – 20 patients:

- 20 patients scanned using a 25-scan free-breathing fast-helical CT (FHFBCT) protocol.
  - Scans registered using pTVreg[1].
- DVF (Deformation Vector Field) and Image Analysis
  - Curls determined by averaging vector fields over four-voxel cubic regions.
    - Smooths short-range curls to focus on long-range behavior.
  - Calculated HU-differences and curls for 24 deformed lung scans.
- Visual inspection of DIR quality; ranked from 1 (high-quality) to 4 (low-quality).
- Pearson correlation between metrics; Point-Biserial correlation with manual evaluation.

Subsequent Analysis:

- 194 patients, including the original 20, underwent same acquisition with deeds registration[2,3].
  - With and without subsequent pTV registration.
- HU-differences used as quality metric.

RESULTS

The Pearson correlation between curl and HU-difference was 0.86. Correlation between the HU difference range and curl threshold volumes was 0.92. Point-Biserial correlations between the manual ratings and each metric were 0.72, 0.67, 0.68, and 0.74 for the HU difference, curl, percent volume within the HU range, and percent volume above the curl threshold, respectively.

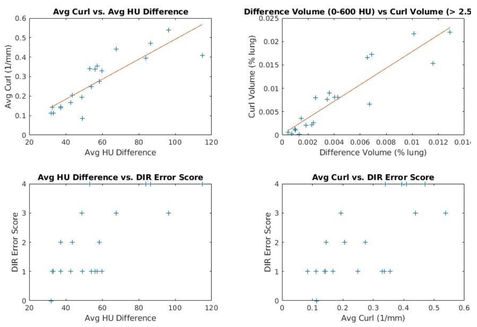


Figure 1: *High correlation is shown between average curl and average HU difference (top left), as well as between the volume of lung with an absolute HU difference between 0 and 600 HU and the volume of lung with a curl above 2.5 mm^{-1} (top right). There is also a clear positive correlation between average HU difference and manual error score (bottom left) and between average curl and manual error score (bottom right).*

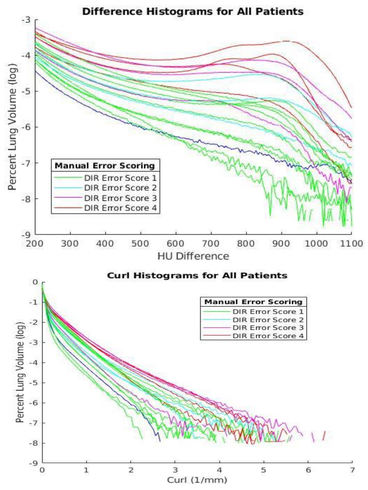


Figure 2: *Absolute HU difference (top) and absolute curl (bottom) are shown against percent lung volume on a logarithmic scale. Each line represents a different patient, and the line colors represent the DIR quality rating given by visual inspection. Generally, poor quality registrations have higher HU differences, particularly at values associated with the misregistration of soft tissues, and higher curls.*

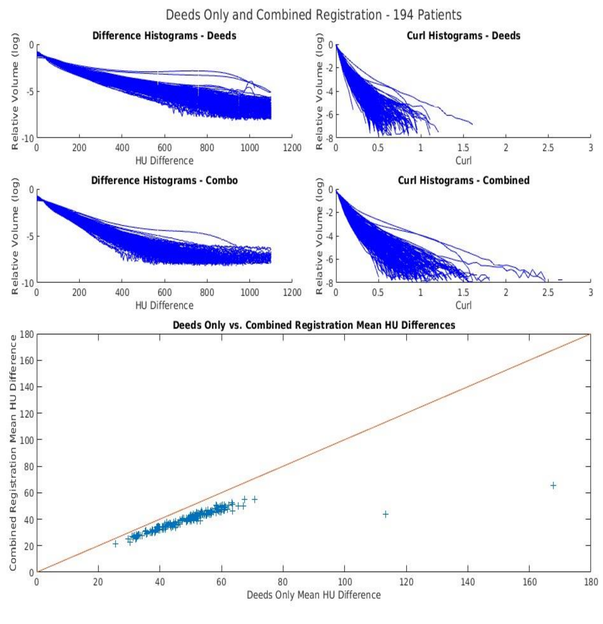


Figure 3: *The same curves from **figure 2** are generated for 194 patients that had their image sets registered with the deeds algorithm (top) and with a combination of deeds and a follow-up pTV registration (middle). These patients did not undergo a manual inspection. These plots indicate that the deeds algorithm performs, on average, significantly better in terms of HU differences when compared to the pTV-only patients. Because of the way deeds works, little or no curl exists in the DVFs. Adding a follow-up pTV registration further improves the HU differences for all patients, while introducing negligible curl. The final plot (bottom) shows the mean HU difference for all 194 patients using each type of registration, with the red line representing unity. The mean HU difference decrease for all patients when the follow-up pTV registration is added and decreases significantly for outlier patients that had large errors present after the deeds registration.*

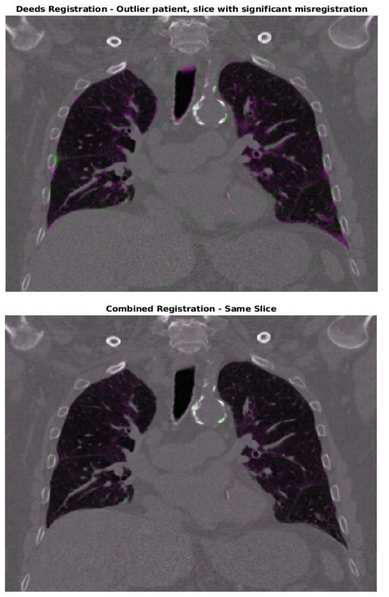


Figure 4: *The above images are generated by overlaying the reference image with the worst registered image for an outlier patient that had particularly inaccurate registration with the deeds algorithm. The purple and green regions indicate areas of HU difference. In the deeds overlap (top), large differences and large regions can be seen throughout the lung. In the combined registration overlay (bottom), these differences are largely mitigated, indicating the efficacy of the combined method in improving deeds registration for outlier patients.*

CONCLUSIONS

Our results indicate that overall DIR quality for pTV registration can be evaluated both by analyzing HU-differences and DVF curl. These metrics exhibit different sensitivity in evaluating different regions: large HU-differences often highlight regions where large soft tissue structures have been misregistered, particularly near the diaphragm; large curl values often indicate non-physical DVF distortion in parenchymal regions that might not otherwise have large HU-differences. Applying these metrics to 194 deeds and deeds with pTV registered patients finds that adding a follow-up pTV registration generally slightly lowers the average HU differences, sometimes greatly lowering the differences in cases with particularly poor deeds registration. Curl was found to be a poor metric for determining deeds registration quality, as deeds allows for little to no curling in its DVFs.

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CONTACT INFORMATION

Email: randosca@mednet.ucla.edu